

Econometrics I

Bonus: How I work?

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Since this is a class about **research methodology** it might be worthwhile to review what my workflow looks like:

- ▶ There are lots of ways to organize your econometric analysis
- ▶ This is just the way that I do things

My Computer

- ▶ I use mostly very standard Mac hardware (a Macbook Air and a Mac Mini)
 - You will be fine with a PC, but I like the Mac's UNIX filesystem and command line.
- ▶ If I need something more than that, I use NYU's HPC cluster, now called **Torch** (it replaced Greene): https://services.rt.nyu.edu/docs/hpc/getting_started/intro/
- ▶ I write mostly Python using VS Code <https://code.visualstudio.com>
- ▶ I use **Claude Code** inside VS Code as a coding assistant
 - Great for debugging, refactoring, plots/tables, and learning unfamiliar packages.
 - You are still responsible for every number: read the code it writes, and ask it to show you checks.
- ▶ I don't recommend using conda or pip for managing Python, I recommend using uv instead.
- ▶ I rarely download software from the web, when I can install it in homebrew <https://brew.sh>
 - When I mean everything, I mean everything (R, Rstudio, Github, Spotify, LaTeX)
 - Also for missing libraries that R links to.

You need to use it!

- ▶ You will probably ignore this advice until something goes horribly wrong.
- ▶ Your projects should probably live in a **private** GitHub repository.
- ▶ If GitHub is annoying you can use the Github Desktop interface
<https://desktop.github.com/download/>.
- ▶ At the very least use Dropbox and pay for the **extended history**
 - But putting Github and Dropbox on the same folder will lead to disaster !
- ▶ You will want to roll back to an older version more than you think!
- ▶ You will want to know exactly what changes your coauthor made all of the time!

Organizing Projects

- ▶ Always separate **code** from **data**
- ▶ Further separate code which **processes data** from code which **analyzes data**.
- ▶ Ultimately you will need a single script that runs the entire analysis from start to finish
- ▶ Start with a folder of **raw data** and a folder of **processed data**.
 - Include directions on exactly where to get the data and **appropriate citations** (including DOI).

chrisconlon Create README.html		c430d4c · 3 years ago	🕒 82 Commits
code	AEA data editor revision	5 years ago	
data	Add design file	5 years ago	
figures	Empty directory Git fix	5 years ago	
tables	Empty directory Git fix	5 years ago	
.gitattributes	Initial commit	5 years ago	
.gitignore	Update .gitignore	5 years ago	
LICENSE	Initial commit	5 years ago	
README.html	Create README.html	3 years ago	
README.md	Fixed Data Citation	5 years ago	
README.pdf	Fixed Data Citation	5 years ago	
requirements.txt	Added PDF readme	5 years ago	
run_all.bat	Create run_all.bat	5 years ago	
run_all.sh	Clean up table code and run file	5 years ago	
wrds_constituents.pdf	Added constituents tutorial to ignore	5 years ago	

Organizing Projects

- ▶ Number your files
- ▶ Try to avoid mixing cleaning/filtering data with **analysis**, so that **all analysis starts from a common dataset** with minimal processing.
 - Otherwise you end up with red flags like tables with different numbers of observations (!)

Let's take a look at <https://github.com/chrisconlon/CommonOwnerReplication>



```
utilities
├── 1_Download_WRDS_Data.py
├── 2_Process_WRDS_Data.py
├── 3_Calculate_Kappas.py
├── firminfo.py
├── investors.py
├── kappas.py
├── our_plot_config.py
├── plots10_kappa_comparison_appendix.py
├── plots11_profit_simulations.py
├── plots1_basic_descriptives.py
├── plots2_kappa_official.py
├── plots3_big_three_four.py
├── plots4_investor_similarity.py
├── plots5_airlines_cereal.py
├── plots6_sole_vs_shared.py
├── plots7_short_interest_coverage.py
└── plots8_individual_firm_coverage.py
```

Good Code has

- ▶ Clear Comment describing what the file does
- ▶ A list of all required **inputs** and **outputs**
- ▶ Detailed comments whenever you merge/drop observations
 - Write separate checks to go through missings/drops/etc.
 - I often just write them to an Excel file (!) to flip through more easily.
- ▶ Shared routines and complicated logic can live in a separate file.
- ▶ Minimal side effects and dependencies.

Avoid Bad Code

```
gen VALID = cond (V = 1R3, 1R6, 2R3, 2R6)
replace VALID = . if SP == 0000
replace VALID = 0 if else
replace VALID = 1 if (SP ==2221 | SP ==2231 | SP==2211 | SP==2311 | SP==2111 | SP==2223| SP==3111)
replace VALID = 2 if (SP==2233 | SP==2331 | SP==3311)
replace VALID = 3 if (SP==2333 | SP==3331)
```

- ▶ Big Blocks of “Replace IF” should probably be a .csv file and a merge.
- ▶ If you can take a big block of changes/drops/etc and make it **data** instead of a code block, do that!
- ▶ This is easier to modify/maintain and catch mistakes.

You may find yourself starting at terabytes of `.csv.gz` files.

- ▶ Goal: filter/drop the rows you don't need and save them to a `.parquet` as quickly as possible.
 - These compress really well and load very fast.
- ▶ You can save multiple `.parquet` files in one directory and treat them as a giant (larger than RAM) file.
- ▶ In python you can use `polars` instead of `pandas` to read files larger than memory.
- ▶ The industrial strength version is `arrow` which works for both R and Python.
 - You can read parts of files and process them chunk by chunk.
 - You can filter without loading the whole file
 - And save directly to `.parquet`
- ▶ NEVER USE `.RDATA` (you will thank me later).

See my project that ingests 2TB of data and saves about 1.5GB of cleaned data in <5min.

You have some choices here

- ▶ Most people use Overleaf for academic papers www.overleaf.com (which NYU subscribes to)
 - The main feature is that multiple people can edit paper at once
 - Otherwise you should use Github/ Version Control
- ▶ You probably want a $\text{\LaTeX}$ install on your computer just in case.
- ▶ Writing a paper in an environment without a spellcheck, etc. is insane (don't do this).
- ▶ Put your citations in a bibliography manager like www.zotero.com otherwise you will spend forever formatting and updating them.
- ▶ Don't worry too much about formatting paper/citations until paper is accepted, but you need them to be **accurate** always.
- ▶ If you are really fancy you can sync all of the tables figures to update the paper automatically (this can lead to surprises).

Thanks!
